

Stakeholder Engagement Plan

Project: Optimizing RAG Document Retrieval for Agronomic Advice

1. Stakeholder Group 1: African Smallholder Farmers

- **Who they are:** Rural agricultural practitioners across Sub-Saharan Africa who manage small plots of land and rely on crop yields for their livelihood and local food security.
- **Why they are important:** They are the ultimate end-users whose localized, colloquial queries (e.g., describing symptoms like "yellowing lower leaves") dictate the input variables the retrieval pipeline must understand.
- **What matters most to them:** Rapid, easily understandable, and highly accurate interventions to save failing crops from pests, diseases, or climate stress.
- **Potential Benefits & Risks:**
 - a. *Benefits:* Timely interventions that protect crop yields and foster agricultural resilience.
 - b. *Risks:* Misdiagnosis (e.g., the AI confusing a potassium deficiency for a nitrogen deficiency) leading to incorrect treatments and subsequent crop failure.
 - c. *Considerations:* The linguistic gap between formal agricultural terminology and local dialects.
- **Engagement Method: Participatory Design**
 - a. *Why it is appropriate:* It allows us to build the system's language comprehension alongside the people who will actually speak to it.
 - b. *How it will be conducted:* We will hold interactive sessions where farmers describe specific crop diseases in their own words. These colloquial phrases will be mapped directly to the formal technical factsheets in our corpus to test and tune our semantic vector representations.
- **Influence on Project Design:** This feedback will directly refine our dataset by exposing the hybrid retrieval pipeline (BM25 + dense embeddings) to a wider variety of informal query structures, improving the model's ability to handle vocabulary mismatch.

2. Stakeholder Group 2: Agricultural Extension Workers

- **Who they are:** Frontline agricultural advisors and government/NGO agents who travel to rural farms to provide expert guidance and diagnose crop issues.
- **Why they are important:** They act as the primary operators of the RAG system in the field. Their hardware limitations dictate our technical constraints.
- **What matters most to them:** Having a fast, reliable, and resource-conscious tool that works on standard low-cost CPU hardware without requiring cloud GPU infrastructure.
- **Potential Benefits & Risks:**
 - a. *Benefits:* The ability to instantly retrieve the exact expert factsheet needed from a 695-document corpus while standing in a field.
 - b. *Risks:* System failure due to high latency, or the pipeline surfacing "hard negatives" (e.g., retrieving a prevention manual instead of an active treatment guide).
- **Engagement Method: User Committees**
 - a. *Why it is appropriate:* Extension workers possess the technical baseline to evaluate

system accuracy and the field experience to evaluate deployment feasibility.

- b. *How it will be conducted:* A representative group of extension workers will be given access to the CPU-only retrieval prototype. They will test the system's speed and evaluate the top-5 retrieved documents (nDCG@5) against real-world scenarios they face.
- **Influence on Project Design:** Feedback from the committee will directly influence our evaluation metrics and deployment decisions. If the cross-encoder re-ranker is too slow on their edge devices, we will adjust the model architecture to strictly prioritize resource-conscious, low-latency execution. Their agronomic expertise will also help us tune the margin penalties for hard negatives.